

OBASI-SAM OTEI

Backend Software Engineer · Python · Django · PostgreSQL

Calabar, Nigeria | 09161101443 | obasonz61@gmail.com

[|github.com/obasi-1](https://github.com/obasi-1) | [| portfolio-frontend-jet-one.vercel.app](https://portfolio-frontend-jet-one.vercel.app)

PROFESSIONAL SUMMARY

Aspiring Backend Software Engineer and Chemical Engineering student with hands-on experience in Python, Django, RESTful API development, relational databases, and backend system design. Skilled in building structured and scalable applications using Django, SQL, PostgreSQL/MySQL, and Object-Relational Mapping (ORM). Combines engineering problem-solving skills with software development knowledge to build efficient, reliable solutions for real-world applications. Seeking opportunities as a Software Engineering Intern, Graduate Software Engineer, Junior Backend Developer, or Entry-Level Software Engineer.

TECHNICAL SKILLS

Programming & Backend: Python, Django, Django REST Framework, RESTful APIs

Databases: SQL, PostgreSQL, MySQL, Database Design, Database Normalization, Object-Relational Mapping (ORM)

Authentication & Security: JWT Authentication, Password Hashing, Token Management, Role-Based Access Control (RBAC)

DevOps & Deployment: Git, GitHub, GitHub Actions, CI/CD, Docker, Render, Vercel, PythonAnywhere

Tools & Practices: Postman, API Testing, Version Control, Agile Methodologies

Engineering & Analytics: UniSim, Process Simulation, Workflow Optimization, Data Modeling

SOFTWARE ENGINEERING PROJECTS

Telemedicine Clinic API

Python | Django | Django REST Framework | PostgreSQL

In Progress

- Developing the backend infrastructure for an online healthcare and appointment management platform.
- Designing relational database models to manage patients, healthcare professionals, appointments, and consultation schedules.
- Building RESTful API endpoints for user registration, authentication, profile management, and appointment booking.
- Implementing data validation and structured relationships to maintain data integrity and support reliable application workflows.

GitHub: [Project Repository] | **Live API:** [Deployment Link]

Industrial Inventory Management System

Python | Django | REST API | SQL

In Progress

- Developing a backend inventory management system for tracking warehouse stock, equipment, and supply chain activities.
- Designing database models and API endpoints to manage inventory records, stock movements, and equipment logs.

- Using Django ORM and SQL queries to support inventory monitoring, reporting, and stock management.
- Applying knowledge of industrial workflows and manufacturing processes to model real-world operational requirements.

GitHub: [Project Repository] | **Live Demo/API:** [Deployment Link]

Role-Based Authentication Service

Python | Django REST Framework | JWT

- Building a backend authentication service to manage secure user access across applications.
- Implementing JWT-based authentication, password hashing, token management, and token refresh functionality.
- Designing Role-Based Access Control (RBAC) to manage permissions for standard users, administrators, and system managers.
- Developing secure API endpoints for user registration, login, authentication, and access management.

GitHub: [Project Repository]

PROFESSIONAL EXPERIENCE

Industrial Trainee

Marian Table Water | Calabar, Nigeria

- Analyzed operational workflows involved in commercial water production and manufacturing.
- Documented production processes, quality control procedures, and key stages of plant operations.
- Prepared and delivered a technical presentation on plant operations, quality control, and production efficiency.

- Identified workflow bottlenecks and areas that could benefit from improved monitoring and automation.

EDUCATION

University of Calabar | Calabar, Nigeria

Bachelor of Engineering (B.Eng.) in Chemical Engineering

In Progress

Relevant Coursework: Process Simulation (UniSim), Fluid Mechanics, Heat Exchanger Modeling

Research: Authored an abstract titled “*AI and Big Data in Oil and Gas Operations*” for the 2026 SPE Nigeria Council Student Paper Contest.

ALX Africa

Software Engineering Programme – ProDev Backend Specialization

Completed

- Completed an intensive software engineering programme focused on backend development, system architecture, database design, API development, and modern software development practices.
- Developed practical knowledge of backend engineering, application architecture, version control, databases, and software development workflows.

ADDITIONAL INFORMATION

Career Interests: Backend Engineering, API Development, Database Systems, Software Architecture, Industrial Automation, and AI-Powered Applications.

Availability: Open to Software Engineering Internships, Graduate Opportunities, Junior Backend Developer Roles, and Entry-Level Software Engineering Positions.